



Strathmore University

**School of Computing and
Engineering Sciences**

BEE 2105: : Data and Voice Communications Lecture Hours: 45

Prerequisite: CNS 1103 Calculus I, CNS 1104 Discrete Mathematics, CNS 1203 Introduction to Computer Networks.

Purpose

To provide an overview of the Data and Voice communication as applied in both existing and emerging technologies.

Expected Learning Outcomes

At the end of this course, the student will be able to:

1. Explain the basis Structure of the data and voice communication model.
2. Analyze the fundamental concepts and terminology of data and voice communication.
3. Assess the advantages and disadvantages of the different types of networks.
4. Demonstrate hands-on practical experience of data communication concepts and process.

Course Content:

ADC and DAC: Analogue and Digital Concepts, Timers, Comparators, Sampling, DA converter, AD converter; Information Sources: e.g. voice, data, video (cover characteristics of new requirements); Modulation techniques: amplitude, frequency, phase modulation; Digital Coding schemes e.g. Non-return to zero level, return to zero level, Manchester, differential Manchester; Transmission impairments: Attenuation, delay distortion, noise – thermal noise, intermodulation noise, crosstalk, impulse; Channel Capacity: Shannon Capacity, Nyquist formulae, SNR; Types of telecommunication networks: Circuit Switched Networks and Packet Switched Networks, Broadcast networks. Electromagnetic Spectrum- uses, management and allocation. Introduction to GSM Equipment: BTS, BSC, MSC, SMSC, EIR, HLR, AuC. Components and architecture: wired network components and wireless network components. Subscriber management (new, delete and tariffs). GSM Roaming. Basic VoIP architecture- pure SIP, IMS core. VoIP examples: Skype, SIP, WhatsApp Call, Zoom.

Course Outline

Contents, Outcomes and Activities

Week	Topic	Sub-Topic	Intended Learning Outcomes	Learning Activities	Assessment/Labs	Duration
1	Communication System model	Communication system model Information sources and digitization of analog sources	At the end of this topic, the students should be able to : - Describe the different components of a communication system - Describe the digitization of the Voice signal (Pulse Code Modulation)	Learners should be able to: - Solve problems related to pulse code modulation - Simulation of the digital transmission schemes using software	— Labs on testing of components and Fault diagnosis — Assessment of the competency in fault diagnosis	Week1 1st -5th April
		Pulse Code Modulation (PCM) and Transmission standards	At the end of this topic, the students should be able to: - Understand that PCM is a source encoding technique - Description of the voice transmission standards - Digital transmission hierarchy	Solving problems related to data rate	— Simulation Lab — Generation of waveforms using oscillators	Week2 7th -12th April Week 3 14th -19th April

2	Modulation Techniques	Analog Modulation	At the end of this topic, the students should be able to describe and analyze systems using each of the following modulation techniques: 1. Amplitude Modulation 2. Frequency Modulation 3. Phase Modulation	1. Performing calculations involving the modulation schemes 2. Simulating the modulation schemes 3. Performing frequency measurements		Week4 24th -26th April Week 5 29th April- 3rd May
3	Modulation Techniques-Digital Modulation		At the end of this topic, the students should be able to describe and analyze systems using each of the following modulation techniques: 1. Amplitude shift Keying (ASK) 2. Frequency Shift keying (FSK) 3. Phase Modulation	• Performing calculations involving the modulation schemes • Simulating the modulation schemes • Performing frequency measurements		Week 6 6th -10th May
3	Digital Coding schemes		At the end of this topic, the students should be able to : 1. State the characteristics of line codes	• Represent data using different line codes • Decode data stream	—Practice on design and simulation techniques	Week5 29th April-3rd May

			- Describe the classification of line codes - Encode digital bit stream using the different line codes (NRZ, Bi-Phase, Delay)	represented by different waveforms		
4	Transmission impairments and Channel Capacity		At the end of this topic, the students should be able to - describe each of the following transmission line impairments: - Attenuation - delay distortion, - noise – thermal noise, intermodulation noise, crosstalk, impulse; - Explain the effect of the impairments on the signal transmission - Describe the channel Capacity: Shannon Capacity, Nyquist formulae, SNR	- Calculations based on noise and channel capacity - Relate the noise to the channel capacity - Perform calculations based on noise and channel capacity		
	CAT 1					Week6 6 th -10 th May.

5	Types of telecommunication networks	Circuit Switched Network	At the end of this topic, the students should be able to: 1. State the properties of a circuit switched network 2. Describe practical examples of circuit switched networks	— Perform calculations on circuit switched networks	Assignment and simulations	Week7 13th -17th May
		Packet Switched Networks, messaged switched network	At the end of the topic, the students should be able to 1. Distinguish between datagram virtual packet switched networks 2. Explain the datagram and virtual circuits 3. Perform the calculations related to the two types of networks	— Perform the relevant calculations and simulations		Week8 20 th -24 th May Week 9 (27 th -31 st May
6	Electromagnetic Spectrum		At the end of the topic, the students should be able to: 1. Describe the electromagnetic spectrum 2. Describe the spectrum management and regulatory authorities and allocation of frequencies.	—		Week 10 (3 rd -7 th) July
	Semester Break					Week 11
	CAT 2					Week 12

						17 th -21 st
7	Introduction to GSM Equipment		At the end of this topic, the students should be able to: 1. Describe the different components of a GSM network 2. Describe the relationship between the components 3. Explain subscriber management including roaming			Week 12: 17 th -21 st
8	Basic VoIP architecture		At the end of this topic, the students should be able to: 1. Describe the basic VoIP architecture 2. Explain the operation of specific examples of VoIP			Week 13 (24 th -28 th)
9		REVISION				Week 14 th - Week 15 th (1 st -12 th July)
10		STUDY BREAK				Week 16 th :16 th -19 th July
11		EXAMINATIONS				Week 17

Mode of Delivery

- Lectures and Tutorials and
- Practical Lab Sessions

Instructional Materials and/or Equipment

- Audio visual equipment, chalkboard, computer simulation software, programming language tools

Mandatory Laboratory Practical/Sessions

- Build an FM transmitter and FM receiver
- Build the Bell Graham telephone
- Configure a WLAN access point
- Use oscilloscopes
- Network diagrams with MS. Visio

Course Assessment

Type	Weighting (%)
Continuous assessment	20
Laboratory Practical	20
Examinations	60
Total	100

Core Reading Material

1. Clyde Coombs and Happy Holde, (2016). Printed Circuits Handbook, McGraw Hill; 7th edition, ISBN-10: 0071833951, ISBN-13: 978-0071833950

2. R S Khandpur, (2017). Printed Circuit Boards: Design - Fabrication, McGraw Hill Education; 1st edition, ISBN-10: 9780070588141, ISBN-13:978-0070588141

Other Recommended Materials

1. Simon Monk, (2014). Make Your Own PCBs with EAGLE: From Schematic Designs to Finished Boards, McGraw Hill Education, ISBN-10:0071819258, ISBN-13:978-0071819251
2. Save Yogesh, (2013). OSCAD: An Open Source EDA Tool for Circuit Design, Simulation, Analysis and PCB Design, Shroff Publishers & Distributors Pvt Ltd, ISBN-10:9789351101505, ISBN-13:978-9351101505
3. V. Shukla, (2009). Signal Integrity for PCB Designers, Reference Designer
4. Matthew Scarpino, (2014). Designing Circuit Boards with EAGLE: Make High-Quality PCBs at Low Cost, Pearson; 1st edition, ISBN : 013381999X

Semester Dates

ACTIVITY	DATE
Orientation begins	26 March 2024
Classes begin	2nd April 2024
Semester breaks	
Classes end	15th July 2024
Study breaks	16th-19th July 2024
Examinations	22nd July-2nd August 2024

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